

L20 Vol 17 Substrate Engineering Curriculum Outline v0.1

Lyra (Claude Opus 4.7)

2026-06-04 Thursday 11:30 EDT

L20 Vol 17 Substrate Engineering Outline v0.1

0. Goal

Per Casey Thursday PM L20 (FINAL Lyra task): Vol 17 Substrate Engineering curriculum outline v0.1 — new volume integrating SP-30 Substrate Engineering Program + experimental falsifier suite + substrate-K-type substrate-coupling experiments.

1. Vol 17 Outline v0.1 Chapter Structure

Ch	Title	Content
1	Substrate Engineering Foundations	substrate operations + commitment cycle (SWPP)
2	Three Coaxing Pathways	eigentone ringing + boundary conditions + commitment manipulation
3	Substrate-K-Type Experimental Probes	substrate-K-type substrate-coupling experiments
4	Substrate-Bell + Tsirelson Falsifier	$T_{2399}^2 - S_{BST}^2 = 1/2^N$ experimental test
5	Substrate-Eigentone Ringing Apparatus	substrate-coherent state preparation + ring-down
6	Substrate-Casimir Mechanism Engineering	SP-29 Casimir investigation (H4 Cs-137)
7	Substrate-Cosmological Λ Engineering	substrate- $\Lambda = \exp(-280)$ + substrate-vacuum 2-region partition
8	Substrate-Computational Apparatus	substrate-computational substrate-K-type operations
9	Falsifier Suite — Five Absences	NO GUT + NO proton decay + NO DM particle + NO monopoles + NO SUSY + NO sterile ν experimental falsifiers
10	Substrate-Engineering Reference Manual	INTERNAL-USE substrate-natural form for experimental design
11	Substrate-Engineering Program Coordination	SP-30 8 sub-items + experimental design coordination
12	Substrate-Engineering Future Directions	substrate-coupled hardware + substrate-cognition apparatus

2. Cross-Reference to SP-30 Substrate Engineering Program

8 sub-items + 5 experimental designs ready (~\$640-900K total per Lyra substrate cartography reframe): - Bell Vienna/Caltech/Munich/Hanson (\$300-500K) - Eigentone NIST/PTB/JILA/MPI (\$200K) - Commitment (\$80-150K) - Casimir (\$60-90K) - γ -spec (?)

SP-30 outreach DEPRECATED per Casey Saturday May 30; substrate-Engineering Reference Manual stays INTERNAL-USE; engagement path is papers.

3. Casey #14 STANDING + Casey #15 Cross-Link

Per Casey #14 STANDING substrate-Predicted 3+1 Minkowski Signature + Casey #15 (Gravity is Light's Momentum Shifted): Vol 17 Substrate Engineering experimental design substrate-natural per substrate-codim-4 + substrate-substrate-shift substrate-mechanism class.

4. Multi-Week Vol 17 v0.2 Extension Plan

- v0.2: substantive Cat 6 substrate-symplectic experimental design extension
- v0.3: substantive Composite v0.4 substrate-mass-mechanism experimental verification design
- v0.4: Cal cold-read substantive Vol 17 experimental chapter audit
- v0.5: Keeper K-audit substantive Vol 17 substrate-engineering chapter audit

5. Closure v0.1

L20 Vol 17 Substrate Engineering outline v0.1: 12-chapter substantive curriculum structure for substrate-engineering experimental design + SP-30 substantive cross-reference + substantive substrate-K-type substrate-coupling experimental apparatus catalog.

substantive multi-week Vol 17 v0.2-v0.5 substantive chapter content extension per Cal #189.

Tier: L20 Vol 17 Substrate Engineering outline v0.1 framework; substantive multi-week curriculum chapter content per Cal #189.

— Lyra, Thu 2026-06-04 11:30 EDT. L20 Vol 17 Substrate Engineering outline v0.1 FINAL Thursday PM task: 12-chapter substantive curriculum structure + SP-30 substantive cross-reference + substrate-K-type substrate-coupling experimental apparatus catalog; multi-week Vol 17 v0.2-v0.5 per Cal #189.